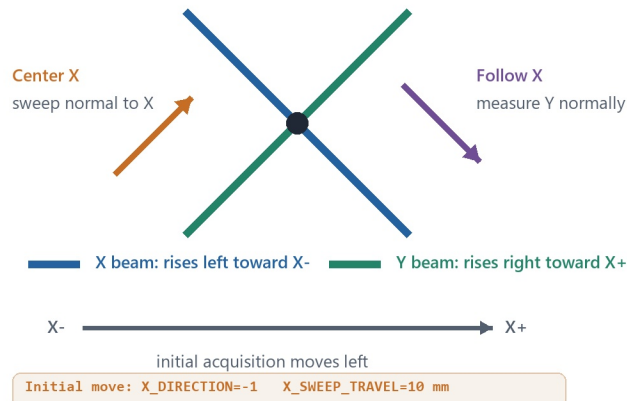


TCP Optical-Cross Calibration

Common-Z diagonal centering, variables, and dual-beam verification

Fixture orientation and 45-degree geometry



Motion and measurement

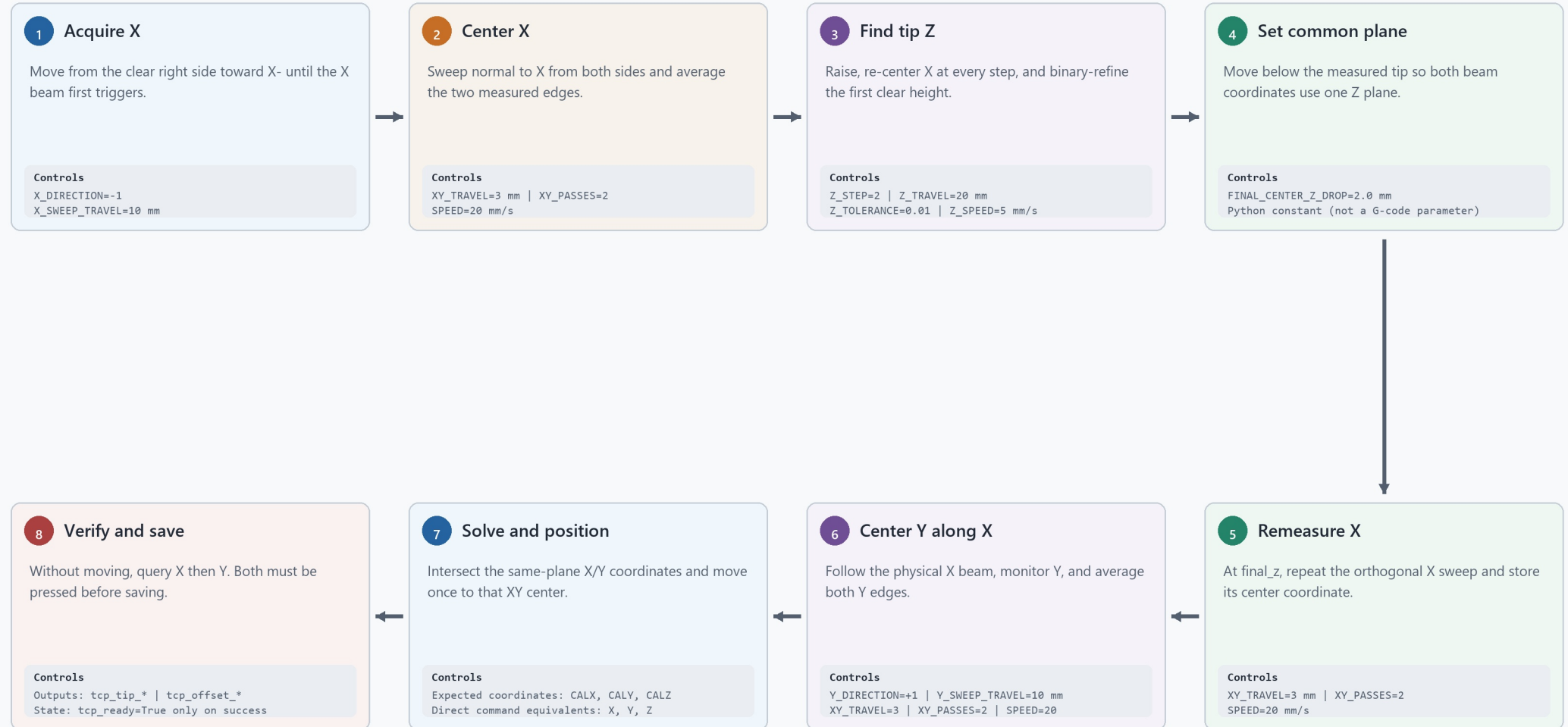
- 1 Start clear on the right/X+ side; move X- until X triggers.
- 2 Track tip Z with bidirectional sweeps normal to the X beam.
- 3 At tip_z - 2 mm, re-center X and follow X while monitoring Y.
- 4 Average both Y edges and solve the same-plane intersection.
- 5 At the stopped position, query X and then Y without movement.

Inputs: TCPSTART X/Y/Z = cclear start; CALX/CALY/CALZ = expected tip; HEIGHT = low search Z

Success: stationary X=PRESSED and Y=PRESSED, then tcp_ready=True

Defaults shown below are the effective values from tcp_calibration.cfg.

Measurement flow and controlling variables



Aliases: RADIUS = XY_TRAVEL; PASSES = XY_PASSES. Command parameters override config defaults for that run.

Failure behavior: tcp_ready is cleared before motion and remains false after any trigger, motion, or verification failure.